MALAPPURAM SAHODAYA APTITUDE TEST 2024-25

SCHOOL LEVEL

Class: 12 PCB **Date:16/11/2024** **Marks: 100**

Name of the Student: ____

GENERAL INSTRUCTIONS

1. ***Read the questions carefully.***
2. ***Attempt all the questions.***
3. ***No negative marking.***
4. ***Answers to be marked in the OMR sheet. Darken the bubble completely. Partially darkened bubbles will not be considered.***
5. ***Question numbers 1 to 100 carries 1mark each.***

Read the following passage carefully and answer the questions:
"The dichotomy between economic growth and environmental sustainability has sparked intense debate. Proponents of growth argue that it is essential for development, while environmentalists contend that it comes at a devastating cost. The notion that these two goals are mutually exclusive is being challenged by innovative solutions. For instance, renewable energy sources and sustainable practices are being integrated into business models."

1. What is the central theme of the passage?
A) The importance of economic growth
B) The impact of environmental degradation
C) The balance between growth and sustainability
D) The role of technology in development

2. What do environmentalists argue in the consequence of economic growth?
A) Job creation
B) Increased revenue
C) Devastating environmental cost
D) Improved infrastructure

3. What is the challenging notion of mutual exclusivity?
A) Government policies
B) Public awareness
C) Innovative solutions
D) Economic downturn

4. What is being integrated into business models?
A) Fossil fuels
B) Renewable energy sources
C) Traditional practices
D) Economic theories

5. What does the word "dichotomy" mean in the context of the passage?
A) Harmony
B) Conflict
C) Balance
D) Division

6. If she had taken the job offer, she ________________________ in New York.
A) Would live
B) Would have lived
C) Will live
D) Lives

7. The company's profits ________________________ by 20% last quarter.
A) Have increased

B) Have been increased
C) Were increased
D) Are increasing
8. The idiom "cut to the chase" means ________________________.
A) To hurry up
B) To get to the point
C) To change the subject
D) To apologize
9. Choose the correct preposition: The new policy applies _______ all employees.
A) With
B) To
C) By
D) From
10. Choose a synonym for the word "proponents"
A) Opponents
B) Supporters
C) Critics
D) Experts
11. Fill up suitably: I'm feeling under the weather today, so I think I'll ________.
A) Take it with a grain of salt
B) Call it a day
C) Bend over backwards
D) Burn the midnight oil
12. Fill up suitably: The Company is trying to ________ its financial problems by cutting costs.
A) Sweep under the rug
B) Take with a grain of salt
C) Turn a blind eye
D) Iron out
13. Fill up suitably: She's very talented, but she's also very ________ and doesn't like to share the spotlight.
A) On the same page
B) In the same boat
C) Burning her bridges
D) Hogging the limelight
14. Fill up suitably: I'm trying to ________ my fear of public speaking by practicing every day.
A) Get to the bottom of
B) Take by storm
C) Overcome
D) Let sleeping dogs lie
15. Fill up suitably: I'm trying to ________ my fear of public speaking by practicing every day.
A. Get to the bottom of
B. Take by storm
C. Overcome
D. Let sleeping dogs lie

16. 1. If she ______ (study) harder, she would have passed the exam.
A. Studies
B. Studied
C. Had studied
D. Would study
17. I would attend the party if I ______ (know) about it.
A. Knew
B. Know
C. Had known
D. Would know
18. Which word is a synonym for "skeptical"?
A. Doubtful
B. Hopeful

C. Optimistic
D. Pessimistic

19. Identify the meaning of the idiom "bend over backwards"?
A. A)to help willingly
B. b) To be stubborn
C. c)to give up easily
D. d) to accommodate

20. Fill in the blank with the correct form of the verb:
"By tomorrow evening, I _______ (finish) this project.
A. will finish
B. b)will be finished
C. will have finished
D. would finish

21. I wanted to attend the concert, ______ I couldn't afford the tickets.
A. but
B. although
C. unless
D. whereas

22. I remember that his dad ________ asks about his studies.
A. Seldom
B. Abroad
C. Weekly
D. evenly

23Which of the following words is an adverb?
A. Anxious
B. Authority
C. Well
D. Perform

24 Select the sentence with the appropriate use of adverb of manner
A. He done his work regularly.
B. He has just done his work.
C. He does his work carefully
D. He has done his work on my table

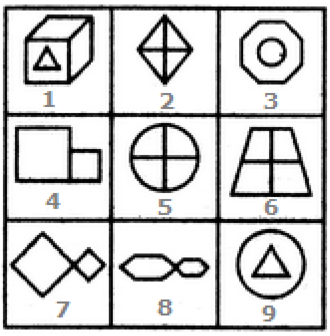
25.Identify the type of Adverb given in bold and mark the correct option.
He drives carefully.
A. Degree
B. Manner
C. Affirmation
D. Reason

26. Which Indian state became the first to pass a resolution to appoint the Chief Minister as Chancellor of all state universities in 2023?
A) Kerala
B) West Bengal
C) Maharashtra
D) Tamil Nadu

27. Which Indian film was selected as the country’s official entry for the 2024 Academy Awards (Oscars)?
A) RRR
B) The Elephant Whisperers
C) Jailer
D) 2018

28. Which space agency successfully launched the Aditya-L1 mission, India’s first mission to study the Sun, in September 2023?
A) NASA
B) ESA
C) Roscosmos
D) ISRO

29. In 2023, India hosted the G20 Summit under which of the following themes?
   A) one earth, one family, one future
   B) Global Unity, Local Progress
   C) Sustainable Development, Global Goals
   D) Peace and Prosperity for All
30. Which Indian city was recently declared a UNESCO World Heritage Site in 2023?
   A) Kolkata
   B) Varanasi
   C) Ahmadabad
   D) Hyderabad
31. In the recent 2023 cricket season, which Indian player became the fastest to score 1,000 runs in T20 internationals?
   A) Virat Kohli
   B) Suryakumar Yadav
   C) Shubman Gill
   D) Hardik Pandya
32. Which city hosted the 18^{th} Asian Games 2023 in which India won its highest-ever tally of medals?
   A) Tokyo
   B) Hangzhou
   C) Seoul
   D) Jakar
33. Which Indian airport ranked first in the world for passenger satisfaction in 2023?
   A) Indira Gandhi International Airport, Delhi
   B) Kempegowda International Airport, Bengaluru
   C) Chhatrapati Shivaji Maharaja International Airport, Mumbai
   D) Rajiv Gandhi International Airport, Hyderabad
34. What was the primary aim of the “Vibrant Villages Programme” launched by the Indian government in 2023?
   A) To develop coastal villages
   B) To enhance border village infrastructure near China
   C) To promote agriculture in rural areas
   D) To create model smart villages
35. India’s new Parliament building, inaugurated in 2023, was designed by which prominent Indian architect?
   A) A)Hafeez Contractor
   B) Bimal Patel
   C) Charles Correa
   D) Bal Krishna Doshi
36. Group the given figures into three classes using each figure only once.



A.1,3,9 ; 2,5,6 ; 4,7,8

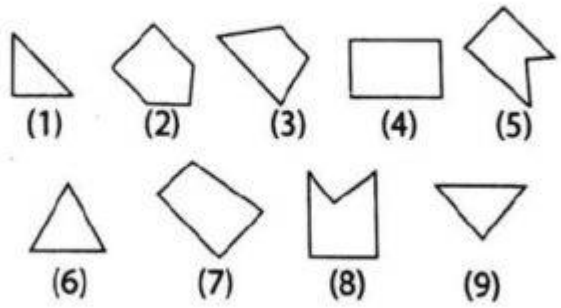
B.1,3,9 ; 2,7,8 ; 4,5,6

C. 1,2,4 ; 3,5,7 ; 6,8,9

D.1,3,6 ; 2,4,8 ; 5,7,9

E. None of these

37. In each of the following question, group the given figures into three classes using each figure only once.



A. (7, 8, 9) (2, 4, 3) (1, 5, 6)

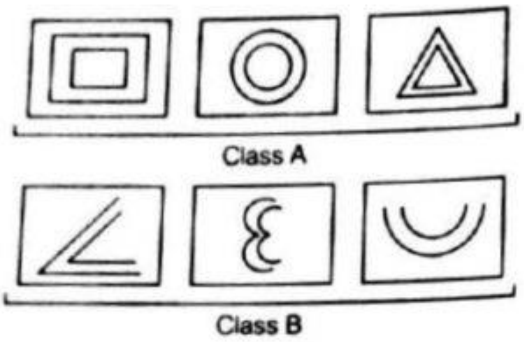
B. (1, 3, 3) (4, 5, 7) (6, 8, 9)

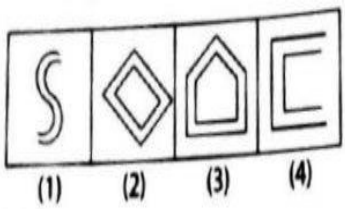
C. (1, 6, 8) (3, 4, 7) (2, 5, 9)

D. (1, 6, 9) (3, 4, 7) (2, 5, 8)

E. none of the above

38. There are two classes of three figures each. Class ‘A’ figures differ in a certain way from the figures in class ‘B’. Which of the four answer figures belong to class ‘A’?





A. Both (1) and (3)

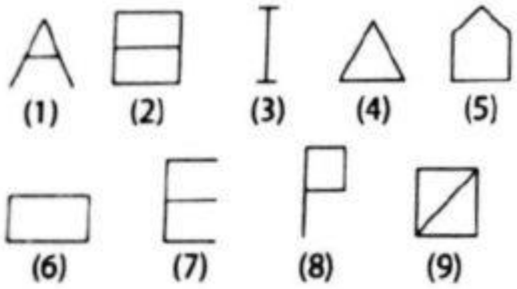
B. (1) and (2) both

C. Both (2) and (4)

D. (2) and (3) only

E. none of the above

39. In the following question, group the given figures into three classes using each figure only once.



A. (1, 3, 4) (2, 5, 9) (6, 7, 8)

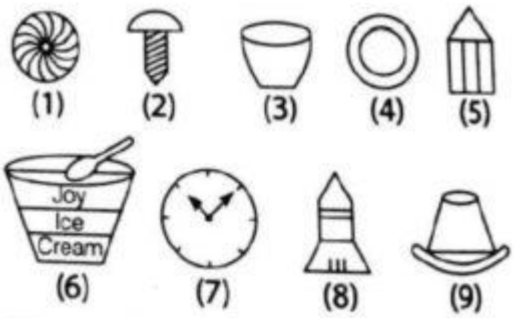
B. (1, 2, 3) (4, 5, 6) (7, 8, 9)

C. (1, 5, 9) (2, 4, 7) (3, 6, 8)

D. (3, 7, 8) (1, 6, 5) (4, 2, 9)

E. none of the above

40. In the following question, group the given figures into three classes using each figure only once.



A. (1, 4, 7) (2, 5, 8) (3, 6, 9)

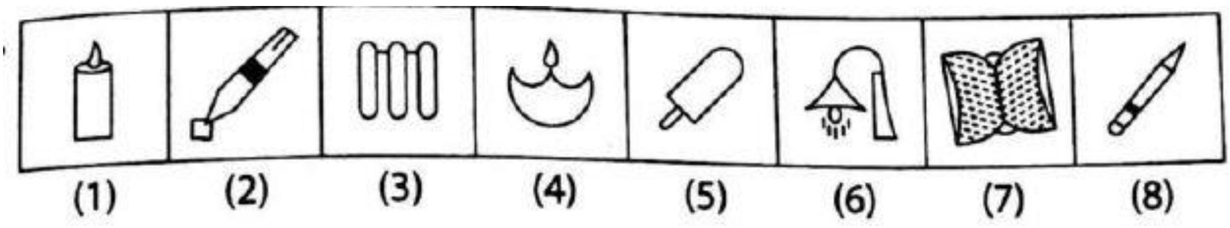
B. (1, 3, 6) (2, 5, 8) (4, 7, 9)

C. (1, 2, 4) (3, 5, 8) (6, 7, 9)

D. (1, 4, 9) (2, 5, 8) (3, 6, 7)

E. none of the above

41. In the below question, a series of figures is given which can be grouped into classes. Select the group into which the figures can be classified from the given responses.



A. (146, 35, 278)

B. (258, 138, 46)

C. (37, 145, 258)

D. (258, 16, 47)

E. none of the above

42. In a row of certain persons, Pradeep is sitting 463 from the left end and 531 from the right end. Find out the total number of persons in that row?

A.963

B. 942

C. 993

D. Can't be determined

E. None of the above

43. In a row all the persons are facing north, Rahul is 33rd from the left end and in the right side of Rahul, there are only 16 persons. Find out total number of person in this queue?

A. 49

B.50

C. 51

D. Can't be determined

E. None of these

44. In a row of 60 persons, Ganesh is 26th from left end. Find out his position from the right end.

A. 35

B. 36

C. 34

D. Can't be determined

E. None of the above

There is a certain relation between two given words on one side of :: and one word is given on another side of :: while another word is to be found from the given alternatives, having the same relation with this word as the given pair has.

45. **Cork: Pop:: Coins : ?**

A. Tinkle
B.Ring
C.Rattle
D. Zoom

E. None of these

46. **Octopus : Mollusca :: Earthworm : ?**

A.Vertebrata
B.Arthropod
C.Platyhelminthes
D. Annelida

E. None of these

47**. Which of the following is the same as Nepal, Bhutan, Myanmar ?**

A. Pakistan
B. Thailand

C. Malaysia
D. Indonesia

E. None of these

48. **Which of the following is the same as Chemistry, Physics, Biology ?**
A. English
B. Science
C. Maths
D. Hindi

E. None of these

49. **Which of the following is the same as LBW, Bowled, Caught?**
A. Six
B. Four
C. Run out
D. No ball

E. None of these

50. **'Warning' is related to 'Danger' in the same way 'Mystery' is related to___________?**
A. Clue
B. Risk
C. History
D. Direction

E.None of these

51. The correct order of reactivity towards the electrophilic substitution of the compounds aniline (I), benzene (II) and nitrobenzene (III) is:
A) II $<$ III $>$ I
B) I $>$ II $>$ III
C) III $>$ II $>$ I
D) II $>$ III $>$ I
52. Order of packing efficiency is :-
(a) HCP $>$ FCC $>$ BCC $>$ simple cubic
(b) HCP $\approx$ FCC $>$ simple cubic $>$ BCC
(c) BCC $>$ HCP $>$ simple cubic $>$ FCC
(d) HCP $\approx$ FCC $>$ BCC $>$ simple cubic
53. In the reaction,

$$BrO_3^-(aq) + 5Br^-(aq) + 6H^+ \rightarrow 3Br_2(\ell) + 3H_2O(l)$$

The rate of appearance of bromine (Br_2) is related to rate of disappearance of bromide ions as following :-
(a) $\frac{d[Br_2]}{dt} = \frac{3}{5}\frac{d[Br^-]}{dt}$
(b) $\frac{d[Br_2]}{dt} = -\frac{3}{5}\frac{d[Br^-]}{dt}$
(c) $\frac{d[Br_2]}{dt} = -\frac{5}{3}\frac{d[Br^-]}{dt}$
(d) $\frac{d[Br_2]}{dt} = \frac{5}{3}\frac{d[Br^-]}{dt}$

54. If in a I^{st^t} order reaction 75% of reaction complete in 4 hrs. Then how much time is required to complete 87.5% of reaction?
(a) 32 h
(b) 6 h

(c) 8 h
(d) 16 h

55. The CFSE for octahedral $[CoCl_6]^4$ is $18000\ cm^{-1}$. The CFSE for tetrahedral $[CoCl_4]^{2-}$ will be :
(a) $18{,}000\ cm^{-1}$
(b) $16{,}000\ cm^{-1}$
(c) $8{,}000\ cm^{-1}$
(d) $20{,}000\ cm^{-1}$

56. Molar conductivities (Λ_m°) at infinite dilution of $NaCl, HCl$ and CH_3COONa are 126.4, 425.9 and $91.0\ S\ cm^2\ mol^{-1}$ respectively. Λ_m° for CH_3COOH will be :-
(a) $290.8\ S\ cm^2\ mol^{-1}$
(b) $390.5\ S\ cm^2\ mol^{-1}$
(c) $425.5\ S\ cm^2\ mol^{-1}$
(d) $180.5\ S\ cm^2\ mol^{-1}$

57. The weight of silver (at $wt. = 108$) displaced by a quantity of electricity which displaces 5600 mL of O_2 at STP will be :-
(a) 5.4 g
(b) 10.8 g
(c) 54.0 g
(d) 108.0 g

58. Which of the following compounds will exhibit cis-trans (geometrical) isomerism ?
(a) 1-Butanol
(b) 2-Butene
(c) 2-Butanol
(d) 2-Butyne

59. Consider the following reaction,

$$\text{ethanol} \overset{PBr_3}{\rightarrow} X \overset{alc.\ KOH}{\rightarrow} Y \underset{H2o}{\overset{H_2SO_4\, room\ temperature}{\rightarrow}} Z;$$

the product Z is :-
(a) CH_3CH_2OH
(b) $CH_2 = CH_2$
(c) $CH_3CH_2 - O - CH_2 - CH_3$
(d) $CH_3 - CH_2 - O - SO_3H$

60. The most suitable reagent for the conversion of $RCH_2OH \longrightarrow RCHO$
(a) $KMnO_4$
(b) $K_2Cr_2O_7$
(c) CrO_3
(d) PCC (Pyridinium chloro chromate)

61. CH_3CHO and $C_6H_5CH_2CHO$ can be distinguished chemically by:
(a) Tollen's reagent test
(b) Fehling solution test
(c) Benedict test
(d) Iodoform test

62. Which one of the following on reduction with $LiAlH_4$ yields a secondary amine
(a) Methyl isocyanide
(b) Acetamide
(c) Methyl cyanide
(d) Nitro ethane

63. Decarboxylation occurs with maximum rate in:
(a) CH_3COOH
(b) $C_6H_5 - COOH$
(c) $C_6H_5CH_2COOH$
(d) CH_3COCH_2COOH

64. Which one of the following is expected to exhibit optical isomerism ? (en = ethylenediamine)
(a) cis $-[Pt(NH_3)_2Cl_2]$
(b) cis $-[Co(en)_2Cl_2]$
(c) trans $-[Co(en)_2Cl_2]$
(d) trans $-[Pt(NH_3)_2Cl_2]$

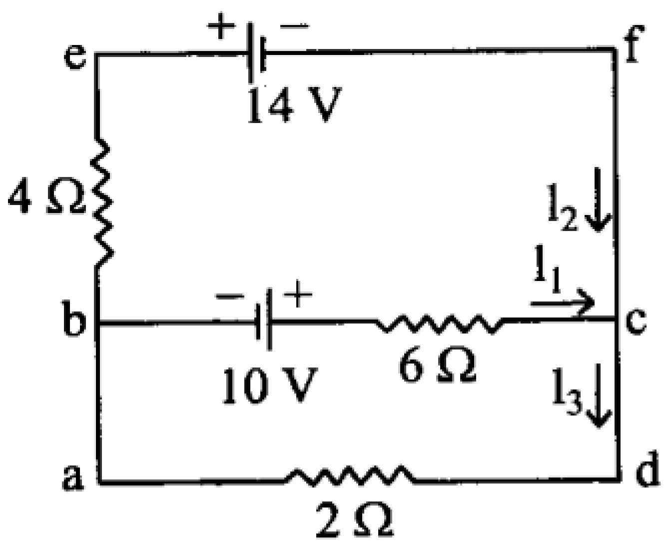
65. Nitrobenzene can be prepared from benzene by using a mixture of conc. HNO_3 and conc. H_2SO_4 In the mixture, nitric acid acts as a/an :
(a) Catalyst
(b) Reducing agent
(c) Acid
(d) Base

66. The electric potential V at any point (x, y, z) in space is given by $V = 4x^2$ volt, where x, y, z are all in metre. The electric field at that point (1 m, 0,2 m) in Vm^{-1} is
a. 8 along the negative x axis
b. 16 along the negative x axis
c. 4 along the positive x axis
d. 4 along the negative x axis

67. In a platinum resistance thermometer, the resistances of the wire at ice point and steam point are of 4Ω and 4.25Ω respectively. When the thermometer is kept in a hot water bath, whose temperature is not known, the resistance of the wire is found to be 4.5Ω. The temperature of the hot water bath is
a. 150°C
b. 100°C
c. 200°C
d. 350°C

68. The distance between two charges $q_1 = +2\mu C$ and $q_2 = +8\mu C$ is 15 cm . Calculate the distance from the charge q_1 to the points on the line segment joining the two charges where the electric field is zero
a. 1 cm
b. 5 cm
c. 3 cm
d. 4 cm

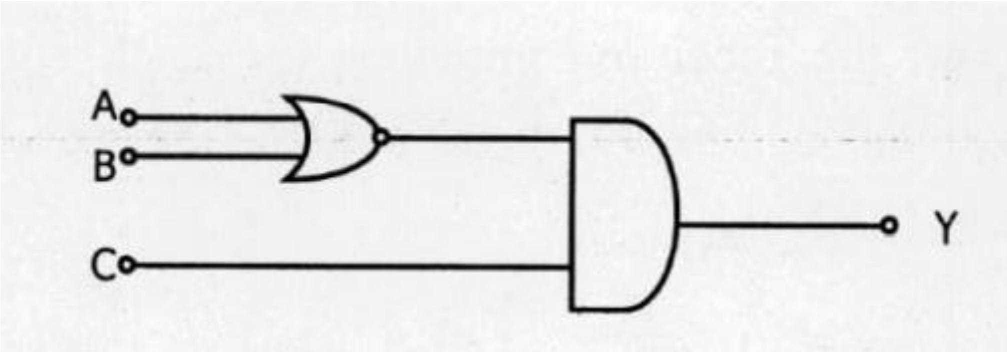
69. The values of the currents I_1, I_2, and I_3 flowing through the circuit given below is



a. $I_1 = -3$ A, $I_2 = 2$ A, $I_3 = -1$ A
b. $I_1 = 2$ A, $I_2 = -3$ A, $I_3 = -1$ A

c. $I_1 = 3$ A, $I_2 = -1$ A, $I_3 = -2$ A
d. $I_1 = 1$ A, $I_2 = -3$ A, $I_3 = -2$ A

70. The inputs A, B and C to be given in order to get an output $Y = 1$ from the following circuit are



a. 0,0,1
b. 1,0,0
c. 1,0,1
d. 1,1,0

71. Eight grams of Cu^{66} undergoes radioactive decay and after 15 minutes only 1 g remains. The half-life, in minutes, is then
a. $15\ln(2)/\ln(8)$
b. $15\ln(8)/\ln(2)$
c. $15/8$
d. $8/15$

72. For a light nucleus, which of the following relation between the atomic number (Z) and mass number (A) is valid
a. $A = Z/2$
b. $Z = A$
c. $Z = A/2$
d. $Z = A^2$

73. What is the minimum thickness (in nm) of a soap film ($n = 1.3$) that results in constructive interference in reflected light if the film is illuminated with light whose wavelength in free space is 620 nm ?
a. 100
b. 120
c. 160
d. 240

74. For any material, if R, T and A represent the reflection coefficient, transparent coefficient, and absorption coefficient respectively, then, for a blackbody which one of the following is true?
a. $R = 1, T = 0, A = 0$
b. $R = 1, T = 1, A = 0$
c. $R = 0, T = 1, A = 1$
d. $R = 0, T = 0, A = 1$

75. When an object is viewed with a light of wavelength 6000Å under a microscope its resolving power is 10^4. The resolving power of the microscope when the same object is viewed with a light of wavelength 4000Å is

a. 10^4
b. 2×10^4
c. 1.5×10^4
d. 3×10^4

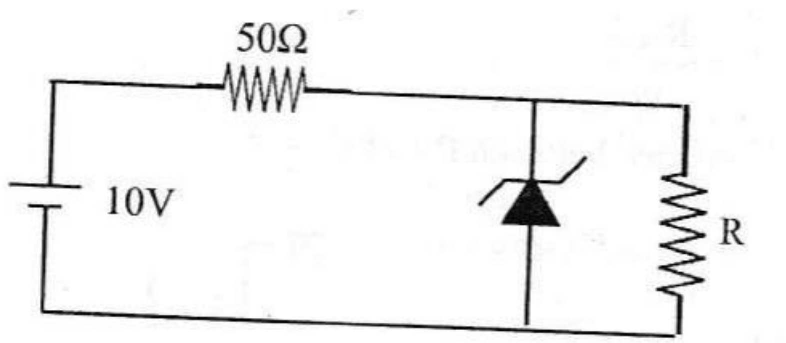
76. Which one of the following is the property of a monochromatic, plane electromagnetic wave in free space?
a. Electric and magnetic fields have a phase difference of $\pi/2$
b. The energy contribution of both electric and magnetic fields is equal
c. The direction of propagation is in the direction of electric filed E
d. The pressure exerted by the wave is the product of energy density and the speed of the wave

77. The electric field of certain radiation is given by the equation E $= 200\{\sin(4\pi \times 10^{10})t + \sin(4\pi \times 10^{15})t\}$ falls on a metal surface having work function 2.0 eV . The maximum kinetic energy (in eV) of the photo electrons is (use Planck's constant $(h) = 6.63 \times 10^{34}$Js and electron charge (e) $= 1.6 \times 10^{-19}$C)
a. 3.3
b. 4.3
c. 6.3
d. 5.3

78. The shunt required to send 10% of the main current through a moving coil galvanometer of resistance 99Ω is
a. 99Ω
b. 11Ω
c. 9Ω
d. 10Ω

79. A lamp consumes only 25% of the peak power in an ac circuit. The phase difference between the applied voltage and the current is
a. $\frac{\pi}{3}$
b. $\frac{\pi}{6}$
c. $\frac{\pi}{4}$
d. $\frac{\pi}{2}$

80. The 6 V Zener diode is shown in figure has negligible resistance and a knee current of 5 mA . The minimum value of R (in Ω) so that the voltage across it does not fall below 6 V is



a. 40
b. 60

c. 72
d. 80

Q.81) Which is the most common type of embryo sac in angiosperms?
a. Tetrasporic with one mitotic stage of divisions
b. Monosporic with three sequential mitotic divisions
c. Monosporic with two sequential mitotic divisions
d. Bisporic with two sequential mitotic divisions

Q.82) Identify the human developmental stage shown as well as the related right place of its occurrence in a normal pregnant woman and select the right option for the two, together.

Developmental stage	Site of occurrence
a. Late morula	- Middle part of Fallopian tube
b. Blastula	- End part of Fallopian tube
c. Blastocyst	- Uterine wall
d. 8-celled morula	- Starting point of Fallopian tube

Q.83) The contraceptive 'Saheli'
a. blocks estrogen receptors in the uterus, preventing eggs from getting implanted
b. increases the concentration of estrogen and prevents ovulation in females
c. is an IUD
d. is a post-coital contraceptive.

Q.84) The plant parts which consist of two generations- one within the other
(1) pollen grains inside the anther
(2) germinated pollen grain with two male gametes
(3) seed inside the fruit
(4) embryo sac inside the ovule
a. (1) only
b. (1), (2), and (3)
c. (3) and (4)
d. (1) and (4).

Q.85) The genotypes of a husband and wife are I^AI^B and I^Ai. Among the blood types of their children, how many different genotypes and phenotypes are possible?
a. 3 genotypes; 4 phenotypes
b. 4 genotypes; 3 phenotypes
c. 4 genotypes; 4 phenotypes
d. 3 genotypes; 3 phenotypes

Q.86) What will be the sequence of mRNA produced by the following stretch of DNA?
3'ATGCATGCATGCATG5'TEMPLATE STRAND

5' TACGTACGTACGTAC3' CODING STRAND
a. 3'AUGCAUGCAUGCAUG5'
b. 5'UACGUACGUACGUAC 3'
c. 3' UACGUACGUACGUAC 5'
d. 5' AUGCAUGCAUGCAUG 3'

Q.87) Which one of the following pairs of items correctly belongs to the category of organs mentioned against it?
a. Nephridia of earthworm and Malpighian tubules of cockroach - excretory organs
b. Wings of honeybee and wings of crow - homologous organs
c. Thorn of Bougainvillea and tendrils of Cucurbita - analogous organs
d. Nictitating membrane and blind spot in human eye - vestigial organs

Q.88) Motile zygote of *Plasmodium* occurs in
a. gut of female *Anopheles*
b. salivary glands of *Anopheles*
c. human RBCS
d. human liver

Q.89) In cloning of cattle, a fertilized egg is taken out of the mother's womb and
a. in the eight-cell stage, cells are separated and cultured until small embryos are formed which are implanted into the womb of other cows
b. in the eight cells stage the individual cells are separated under electrical field for further development in culture media
c. from this upto eight identical twins can be
d. produced (d) the egg is divided into 4 pairs of cells which are implanted into the womb of other cows.

Q.90) Match the following organisms with the products they produce.

(A) *Lactobacillus*	(i) Cheese
(B) *Saccharomyces cerevisiae*	(ii) Curd
(C) *Aspergillus niger*	(iii) Citric acid
(D) *Acetobacter aceti*	(iv) Bread
	(v) Acetic acid

Select the correct option.
a. (A)-(ii), (B)-(i), (C)-(iii), (D)-(v)
b. (A)-(ii), (B)-(iv), (C)-(v), (D)-(iii)
c. (A)-(ii), (B)-(iv), (C)-(iii), (D)-(v)
d. (A)-(iii), (B)-(iv), (C)-(v), (D)-(i)

Q.91) Match List-I with List-II:

	List-I		List-II
A.	Genetically engineered Human Insulin	I	Gene therapy
B.	GM Cotton	II	E. coli
C.	ADA Deficiency	III	Antigen-antibody interaction
D.	ELISA	IV	*Bacillus thuringiensis*

Choose the correct answer from the options given below:

a. A-III, B-II, C-IV, D-I
b. A-II, B-I, C-IV, D-III
c. A-IV, B-III, C-I, D-II
d. A-II, B-IV, C-I, D-III

Q.92) What is the fate of a piece of DNA carrying only gene of interest which is transferred into an alien organism?

A. The piece of DNA would be able to multiply itself independently in the progeny cells of the organism.
B. It may get integrated into the genome of the recipient.
C. It may multiply and be inherited along with the host DNA.
D. The alien piece of DNA is not an integral part of chromosome.
E. It shows ability to replicate.

Choose the correct answer from the options given below:

a. A and B only
b. D and E only
c. B and C only
d. A and E only

Q.93) Match List-I with List-II.

	List-		List-II
A.	Migratory flamingoes and resident fish in South American lakes	I	Interference competition
B.	Abingdon tortoise became extinct after introduction of goats in their habitat	II	Competitive release
C.	Chathamalus expands its distributional range in the absence of Balanus	III	Resource Partitioning
D.	Five closely related species of Warblers feeding in different locations on same tree	IV	Interspecific competition

Choose the correct answer from the options given below:

a. A-I, B-IV, C-III, D-II
b. A-IV, B-I, C-II, D-III
c. A-III, B-I, C-II, D-IV
d. A-II, B-IV, C-III, D-I

Q.94) Match List - I with List - II

	List - I		List - II
(A)	Hydrarch succession	(I)	Gradual change in the species composition
(B)	Xerarch succession	(II)	Faster and climax reached quickly
(C)	Ecological succession	(III)	Lichens to mesic conditions
(D)	Secondary succession	(IV)	Phytoplankton to mesic conditions

Choose the correct answer from the options given below:

a. (A)-(IV), (B)-(II), (C)-(III), (D)-(I)
b. (A)-(III), (B)-(I), (C)-(IV), (D)-(II)
c. (A)-(I), (B)-(IV), (C)-(II), (D)-(III)
d. (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

Q.95) Tropical regions show greatest level of species richness because

A. Tropical latitudes have remained relatively undisturbed for millions of years, hence more time was available for species diversification.
B. Tropical environments are more seasonal.
C. More solar energy is available in tropics.
D. Constant environments promote niche specialization.
E. Tropical environments are constant and predictable.

Choose the correct answer from the options given below.

a. A, C, D and E only
b. A and B only
c. A, B and E only
d. A, B and D only

Q.96) Match List - I with List - II

	List -		List - II
(A)	Deforestation	(I)	Responsible for heating of Earth's surface and atmosphere
(B)	Reforestation	(II)	Conversion of forested areas to non-forested areas
(C)	Green-house effect	(III)	Natural ageing of lake by nutrient enrichment of its water
(D)	Eutrophication	(IV)	Process of restoring a forest that once existed but was removed

Choose the correct answer from the options given below:

a. A-(IV), B-(III), C-(II), D-(I)
b. A-(I), B-(II), C-(III), D-(IV)
c. A-(III), B-(I), C-(II), D-(IV)
d. A-(II), B-(IV), C-(I), D-(III)

Q.97) Which one of the following can be explained on the basis of Mendel's Law of Dominance?

A. Out of one pair of factors one is dominant and the other is recessive.
B. Alleles do not show any expression and both the characters appear as such in F2 generation.
C. Factors occur in pairs in normal diploid plants.
D. The discrete unit controlling a particular character is called factor.
E. The expression of only one of the parental characters is found in a monohybrid cross.

Choose the correct answer from the options given below:

a. A, B and C only
b. A, C, D and E only
c. B, C and D only
d. A, B, C, D and E

Q.98) Identify the set of correct statements:

A. The flowers of Vallisneria are colourful and produce nectar.

B. The flowers of water lily are not pollinated by water.
C. In most of water-pollinated species, the pollen grains are protected from wetting.
D. Pollen grains of some hydrophytes are long and ribbon like.
E. In some hydrophytes, the pollen grains are carried passively inside water.
Choose the correct answer from the options given below.

a. C, D and E only
b. A, B, C and D only
c. A, C, D and E only
d. B, C, D and E only

Q.99) Given below are two statements:
Statement I: Vas deferens receives a duct from seminal vesicle and opens into urethra as the ejaculatory duct.
Statement II: The cavity of the cervix is called cervical canal which along with vagina forms birth canal.
In the light of the above statements, choose the correct answer from the options given below:

a. Both Statement I and Statement II are false.
b. Statement 1 is correct but Statement II is false.
c. Statement I is incorrect but Statement II is true.
d. Both Statement I and Statement II are true.

Q.100) Which of the following statements are correct?
A. Reproductive health refers to total well-being in all aspects of reproduction.
B. Amniocentesis is legally banned for sex determination in India.
C. "Saheli" a new oral contraceptive for females was developed in collaboration with ICMR (New Delhi).
D. Amniocentesis is used to determine genetic disorders and survivability of foetus.
Choose the most appropriate answer from the options given below:

a. (B) and (C) only
b. (D) and (C) only
c. (A), (B) and (D) only
d. (A) and (C) only

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